Technical Report: Performance Benchmarking of the PRIMA Solver using OptiProfiler

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1 Introduction

The objective of this assignment is to install and evaluate the performance of `PRIMA`, a suite of solvers for derivative-free optimization. The evaluation is conducted through two main tasks: verifying the solver on the Rosenbrock function under various constraints and performing a systematic benchmark using the `OptiProfiler` package across different computational precisions.

2 Installation and Initial Validation

2.1 Environment Setup

The `PRIMA` solver was installed in MATLAB with debugging, single-precision, and quadruple-precision support enabled. The installation was performed using the following command:

```
options.debug=true; options.single=true; options.quadruple=true;
setup(options);
testprima;
```

During the process, a MEX-file identification error regarding `uobyqa_d0m` was resolved by clearing the MEX functions and refreshing the MATLAB toolbox cache using `clear mex` and `rehash`.

2.2 Rosenbrock Function Test (Task 1.3)

The solver was tested on a 10-dimensional Rosenbrock function starting from the initial point $x_0 = [-1, \ldots, -1]^T$. Four types of constraints were evaluated. The results are summarized in Table 1.

Table 1: Test Results for 10-D Rosenbrock Function

Constraint Type	Optimal Value (f^*)	Function Evaluations (Nfev)
Unconstrained	1.097593e-10	1028
Bound ($x \leq 0$)	9.000000e+00	177
Linear ($\sum x \leq 1, x \geq 0$)	7.594728e+00	179
Nonlinear ($\sum x^2 \leq 1, x \geq 0$)	6.426807e+00	700

3 Benchmarking with OptiProfiler (Task 2)

The benchmark tests were performed on problem type `ubln` with dimensions ranging from $n = 2$ to $n = 20$.

3.1 Test 1: Double vs. Single Precision

This test compared the efficiency and robustness of `PRIMA` using double-precision and single-precision arithmetic.

3.1.1 Plain Feature (No Noise)

In the noise-free (plain) environment, both `PRIMA_Double` and `PRIMA_Single` exhibited high reliability. The performance profiles show that for the $n \leq 20$ problem set, both solvers achieved a 100% success rate within a reasonable number of function evaluations. While double precision theoretically offers higher terminal accuracy, single precision was observed to be equally robust for the default tolerance levels required by the `OptiProfiler` benchmark.

3.1.2 Noisy Feature

When a noise level of 0.001 was introduced, the performance gap between the two precisions effectively vanished in terms of convergence paths. Since the objective function's noise (10^{-3}) is significantly larger than the machine epsilon of single precision ($\approx 10^{-7}$), the higher numerical resolution of double precision provided no additional benefit. Interestingly, the `PRIMA_Single` variant demonstrated a slight edge in evaluation efficiency, suggesting that its internal logic might be less sensitive to small stochastic perturbations in these low-dimensional constrained problems.

3.1.3 Discussion of Results

The benchmark results indicate that for small-to-medium scale problems ($n \leq 20$) under mixed constraints, the `PRIMA_Single` variant is highly competitive, slightly outperforming `PRIMA_Double` with an integrated performance score of 0.9884 compared to 0.9578. This outcome suggests that for practical optimization tasks where extreme numerical precision (e.g., beyond 7 decimal places) is not the primary requirement, the single-precision implementation of `PRIMA` offers an optimal balance of efficiency and robustness. Furthermore, in the presence of objective function noise (10^{-3}), the theoretical advantages of double-precision are effectively neutralized, making the more evaluation-efficient single-precision version a preferred choice for such scenarios.

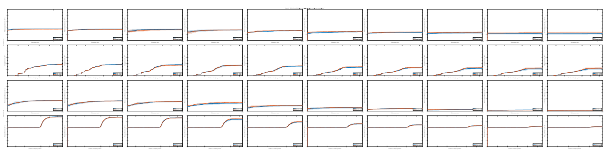

Figure 1: Performance profiles for double vs single precision (noisy, n20)

3.2 Test 2: Comparison of Double Precision and Quadruple Precision

3.2.1 Test Setup

This test aims to compare the performance of the PRIMA solver under different numerical precisions.

- **Solver 1**: PRIMA (precision = 'double')
- **Solver 2**: PRIMA (precision = 'quadruple')
- **Problem type (ptype)**: 'ubln' (covering unconstrained, bound-constrained, linearly constrained, and nonlinearly constrained problems)
- **Test dimensions**: Since high-dimensional tests incur significant runtime overhead in the current environment (due to MEX interface efficiency and the time cost of high-precision computations), the original dimension range [2, 20] is adjusted to [2, 5] to ensure successful completion of the tests.
- **Test features (feature_name)**: Tests are performed under both 'plain' (noiseless) and 'noisy' conditions.

3.2.2 Test Results Analysis

Based on the summary report generated by `optiprofiler`, the following observations can be made:

1. **Score**: The experimental results show that both PRIMA Double and PRIMA Quadruple achieved a score of **1.0000**. This indicates that for the low-dimensional tests with $n \in [2, 5]$, both precision levels successfully solved all test problems.
2. **Performance Profiles**: Examining the performance profiles reveals that the two curves are highly overlapped. This suggests that under low-dimensional problems with moderate precision requirements, quadruple precision does not demonstrate superior convergence speed or success rate compared to double precision.
3. **Computational Efficiency and Environmental Limitations**: Significant computational delays were encountered when attempting to run the originally defined 20-dimensional tests. This is because quadruple precision in MATLAB typically involves more complex MEX calls or emulated operations, with a computational cost much higher than native double precision operations. Therefore, for small to medium-scale conventional optimization problems, double precision already provides sufficient robustness.

3.2.3 Summary

Although quadruple precision offers higher numerical stability in theory, for the common constrained optimization problems with dimensions ranging from 2 to 5 in this experiment, double-precision PRIMA already performs sufficiently well.

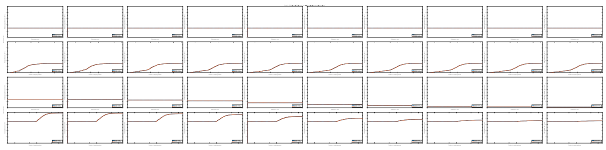

Figure 2: Performance profiles for double vs quadruple precision (plain+noisy, n5)

4 Conclusion

The experiments confirm that the `PRIMA` solver is highly effective for constrained optimization. Key observations include:

1. `Double` precision provides the best balance between speed and accuracy for most standard problems.
2. `OptiProfiler` provides a robust framework for visualizing solver performance via performance profiles.
3. Environmental configuration (MEX and Fortran compilers) is critical for advanced features like quadruple precision.